ActInf Livestream #015.2 ~ "Free-Energy Principle, Computationalism and

Livestream | 1:54:11

hello everyone welcome to
the active inference live stream from
the active inference lab
it is active inference live stream 15.2
today and it is february 9th 2021
welcome to the active inference lab
everyone
we are an experiment in online team
communication
learning and practice related to active
inference you can find us
on our website twitter email youtube
discord or keybase this is a recorded
and an archived live stream
so please provide us with feedback so
that we can improve on our work
all backgrounds and perspectives are
welcome here and we will
follow good video etiquette for live
streams
today in the active inference live
stream you can check out the calendar
for 2021
at this link on the left side and we're
here with the red arrow
on february 9th for 15.2 this is our
second
group discussion with the two authors of
an awesome
paper about realism and computationalism
and instrumentalism
and so this follow-up discussion we're
going to be probably taking it in a lot
of different
directions bringing up some ideas that

we heard about in earlier
15 discussions and sending us off with
momentum
into the future of 16 and beyond
in 15.2 we're going to go around and
hear from everybody
in the introduction section and the
warm-up and then we will
jump around go to some questions that we
had from last week
and anything else that people are
thinking about will be awesome
to hear so for the introduction
and warm up i'm daniel i'm a
postdoctoral researcher in california
and i will pass it
first to shannon
hey i'm shannon i'm a phd student at the
university of california in merced
and i'll pass it to stephen
hello i'm stephen i'm here in toronto
i do a practice based phd and
i will pass it over to dave
okay i'm a retired i.t guy with a
background in
process philosophy and uh natural
language
processing living in the philippines
and let's go to ines
hi everyone so my name is inish i am
based in the berlin school of mind and
brain
and i work in philosophy of cognition
and i'm very happy to be here thanks
daniel pass it to
blue
everyone i'm blue knight i am an
independent research consultant based

out of
new mexico and i will pass it to alex
thanks hi everyone i'm alex i'm in
moscow russia
and i'm a researcher in systems
management school
and i pass it to scott
hi folks i'm scott david i'm the
director of the information risk
research initiative at the university of
washington
applied physics lab i've been a lawyer
for 30 years
and we engineer legal and
socio-technical systems
together thanks is there anyone else to
pass to
daniel thomas
hi everyone um my name is thomas van s
and i work in antwerp on a
philosophy of cognition
cool well this will be a really fun
discussion
so let's just get warmed up with kind of
the pivot point with the phase
transition from
all of our disparate lives whatever we
were thinking about before this
conversation
and get warmed up into the paper and so
the three
warm-up questions are on here anyone who
raises their hand can just
take a stab and it'd be cool to hear
from everyone
what is something you're excited about
today what is something you liked or
remembered about the paper

and then what is something that you're
wondering about or would like to have
resolved
by the end of today's discussion so
while
people are raising their hand one thing
i was curious about is
what leads one to be an instrumentalist
or
a realist like how do we um
make that choice what kind of sends us
down one
road or another or is it something that
switched back and forth
just something i was curious about
anyone else have a um thought on one of
these
warm-up questions or just something that
stayed with them from the paper
um dave and then anyone else who raises
their hands
yeah i would hope that folks don't get
stuck
in having to choose between
instrumentalism
and really getting into the meat of
causation
in any particular topic that you're
interested in uh you gather data
by looking at the envelope of a
phenomenon
and when you've gotten enough data you
just say well what is it like to be one
of them
and uh one hopes you've got enough
mirroring or modeling of the relations
that you can start coming up with some
[Music]

some mirrored explanations of how to
what is it like to be uh
a thermostat
cool thanks anyone else steven
yeah um yeah i think that this
question around thinking about using an
instrument
is quite interesting sort of that
embodied sense of like
i'm probing something with an instrument
um as opposed to maybe using a tool
which maybe is a bit more like i can
hold that in my hand you know
i know what my tool does but my
instrument there's like a little bit of
a gap so i i like that
way of thinking about how we are
working with something between us and
the world and where does the idea of a
system
go in there or where does the idea of a
tool or where does the idea of
you know it actually in some ways is
kind of
realist in the sense of how we try and
think about things physically so i quite
like that
ironically shannon
and then anyone else yeah so i don't
know what it's like to be a thermostat
um but that made me think
of the dynamics of a crowd that consists
of a lot of individual
people and no
one person in the crowd know what knows
what it is like to be the
whole crowd and you can identify certain
behavior that's happening

at the level of the crowd and we can be
super instrumentalists and say this
crowd is
navigating a space or something
um and that sits fine with me but
somehow when i transition to neurons in
a brain i want to say that there is
something realist
and we don't have to be instrumentalist
about the brain we can say
this population of neurons is is
representing some
activity but i don't want to say that
about a crowd
but if you put those two side by side i
should i should have the same view
against both of them
interesting thanks yeah we line up all
these kind of mappings
and then it's almost like um well the
linear regression
is an instrumental um way of doing
statistics so then if we're going to be
serious about instrumentalism
could we replace linear regression with
some other statistical technique or
does it come down to the details of the
instrument in s and then anyone else who
raises their hand
yeah i was thinking about um dave's
remark actually
um which i i also agree with and i think
that
dave's remark allows us to say a little
bit more about
why we need to be careful about being
realist and uh
that is um with the fact that if we're

going to
model say a cognitive phenomenon
say perceptual learning on something or
a phenomenon that is cognitive that
entails also what it is like
[Music]
then probably in modeling that cognitive
phenomena and we are going to have to
take a very realist stance
because you won't be able to capture the
what it is like
so in a way if we are taking the realist
path
and saying that well my model is not
only modeling
the cognitive phenomenon but actually
that's how the phenomenon
works by using these models and these
representations
then we are we also are committed to the
implication
that um that particular phenomenon does
not have or is not permeated by
any phenomenological aspects because we
are applying a reductionist model and
taking a realist perspective
so taking a release perspective is not
going to be useful if we want to account
for the phenomena
the phenomenology aspect of the
cognitive phenomenon
yeah thanks for that
um and anyone else can raise their hand
i have a question for
the authors so the title talks about
computationalism
but one of the main tensions in the
paper is about

instrumentalism and realism but the
title has computationalism and realism
so what is the relationship between
computationalism
and instrumentalism
yeah ns
yeah so um i can say something about
that um
so in computationalism we can take
either instrumentalism or
realism which is cool it's fine um
one way to think about this is that i
can be
uh completely in agreement with
computational
science and use all the apparatus and
machinery that computation
science has to offer at our disposal and
i can use that from an instrumentalist
perspective
now the other way is being realist
within computationalism
and i'm going to say that it's not only
a tool that i use this computational
machinery
but it's also um the way that in this
case the target system which
is the brain or cognitive activity works
so these are the two options of um or
the two uh um options that we have
within the computation within
computationalism
the the typical um the typical
standpoint
is to be a realist about it that's the
history of philosophy of mind
is to apply a realist attitude
towards our scientific models of the

brain or cognitive activity
so it's not only that i have a
representation
or i'm developing a representation of
what i think the brain is doing in
generating
uh these data that i have access to but
i also think that
this representation is also in the brain
that's also how the brain works by
having these sort of representations
permeated with all these cultural
constructs that we have like numbers
logics
uh bayesian inference concepts and all
of that
is in the brain too
cool thanks thomas and then anyone else
yeah um so i think that's that's exactly
right
and um last time it came up as well that
uh it seems like only philosophers are
worrying about this or
making this uh distinction or anything
um
but the thing is that you see that in in
certain scientific projects as well
um when people
uh i've heard anyway
of a research project where people are
trying to identify
particular priors in the brain
are trying to find neural correlates of
the priors in bayesian probability
distributions
um that only
makes sense like the entire research
project only makes sense

if you think the brain is literally
engaging in bayesian infants
so it really matters in that sense also
for
many scientific projects whether you
take an instrumentalist or realist
attitude
thanks yeah i think it's something we
can kind of
tease out and return to is what are the
implications
for where we fall out on this
instrumentalism and realism
continuum or what are the consequences
for different projects or for different
kinds of explanations predictions
models um is it possible to use bayesian
statistics instrumentally on the brain
and be totally neutral and talk about it
like it's a
uh like uh you can model it usefully as
a computer or as doing bayesian
statistics
but then not take the realist turn and
say that it is doing that
or do those two tracks run so close to
each other
that maybe it's a just a slip from one
to the other
and what would be the consequences of
that so stephen and then anyone else
i'd be curious to know whether you've
got the idea that when you
are trying to probe
the brain or try to understand it then
you take an instrumentalist
perspective i was wondering if there is
like a level at which realism

is sort of still seen to be there like
even if it's
in a sense that can't be seen i.e the
free energy principle because it's a
principle
and it's and the idea of non-equilibrium
steady states or the fact that it's
you know that you need the dynamics to
be happening rather than it being
a system that can be just snapshotted
is that in principle form
still coherent to have as like a
relatively
realist interpretation it's just that
doesn't necessarily mean that any of
your models
are realist they're just instruments or
is it not that is there is there
something like that kind of
interaction at a more foundational level
cool question thank you um dave and then
anyone else
yeah on that point i would recommend you
look
at the um mark some
lecture series from 2013 on
the conscious ego he is
both a neuroscientist and a
psychoanalyst
he is also currently collaborating with
carl
fristen and is in like two weeks
bringing out a book
that uh integrates dream
study psychotherapy and
um bayesianism
solm s
cool any other thoughts on the

introduction
or we can um just start with some of
these
questions that we had raised as a group
last week so
yeah sounds good we'll just kind of
i guess walk through a couple of these
questions people can take it in any
direction though so there's not too much
of a linear ordering here but
we'll just walk through them so this
first question um
several people spoke to it last week and
it was related to
conscious awareness and to self
modeling so i um was reminded of this
with steven's question about instrument
and tool
so like do we have an i
is there something that isn't you know
is the uh maybe i'll pick a different
word that's not you know literally the
same e y e or i but let's just say like
um for tactile sensation is it that we
have
proprio sensation or that we are proprio
sensation or do we make a model
of proprio sensation or are we agnostic
on how it really is with proprio
sensation but we're just gonna model it
a certain way there's a lot of
directions that we can
take this but i think that the
interesting question that was raised is
where does the self awareness come from
and then is a self uh
enacting these types of uh
inference schemes or is it as if they're

in acting schemes
how do we think about that for ourselves
how do we think about that for other
systems
um yep ines then chan
um i think that um one point
uh at which we can start this
conversation
is by asking ourselves um what is
modeling what are the motivations
for us to uh get around and
uh construct a model of something
so i would say um that we model things
that
we do not have a complete access to
we are somehow missing something about
you have an object of study
but somehow we are missing um
some part of it that is puzzling to us
and we don't have
a a direct access to it almost like
saying we don't have the true posterior
so we need to find it by um uh
coming up with theories um and uh by
thinking about
uh what would be uh what would make
sense um
to explain that particular phenomenon
that we are
puzzled about so i think this is
important for this question of
self-modeling because then it raises the
question of
whether i have access to the self or not
because if we think that however when we
answer this question is going to be very
relevant to however we think this
question

if we think that we the self is
something that is hidden away
and we don't have a complete access to
it then we could perhaps
think about that oh there's motivation
to model the self
how is a different question if we think
that no i have a direct
access if to anything to the self on a
conscious level
um i might not understand everything
about myself or the
self um but i do have a direct access in
which case
perhaps there is no motivation to model
it um
so yeah i just think this is a nice way
to to start thinking about
this thanks
shannon and then thomas i'm actually
going to hold my comment
until a little bit later all right cool
thomas
yeah um so i think two other
very important questions here
um
i don't know maybe there's another one
um
what is access
um what do we mean
with the conscious level
um
what is the self that we are supposed to
be modeling or not modeling
um
i think that that might be it and and
this seems a little nitpicky
and typically philosophical i can

imagine but it
really is very important that we get
clear on these things because
[Music]
some people will have a very very narrow
definition
of what you have access to um
which will very quickly kind of mean you
will need to
represent or model just about anything
um whereas if you have a broader
definition of what we have access to
that's going to
change the story completely um
the same thing goes with the self some
people have different
varieties of the self there's this sort
of pre-reflexive
self that is just sort of self-hood
some people call it that you just
have a feeling of being yourself and as
an acting person
and there's a narrative self that's sort
of the story you make up about yourself
like what sort of person am i i'm
etc
those are all going to be enormously
important questions to answer before we
can really
talk about what self-modeling on the
conscious level
even is i think
great point it is very philosophical to
to take one question
and just fractalize ask sub questions
but it's
so true it really does matter if we
think that we're

going to explain some consciousness
in one definition then we might be off
base so i guess since these are big
questions what might the free energy
principle or
active inference do to help us resolve
or at least approach these
ideas shannon and then it could be on
that or or anything else
than anyone else yeah so coming up from
what you said do we have proprioception
like do we have a certain sensation or
are we that sensation are we
proprioception
um and anest's question like what are
the motivations for modeling
um and we model if we're somehow missing
the full picture
um and this brings me to like in
yakipovi's
book predictive mind right the brain
doesn't have access to
the sensory environment the brain has
access to
electrical firing and so it's trying to
model
what in the environments could have
caused or what
um what could have caused that certain
pattern of electrical firing
and back to last week when we were
talking about representations having
some sort of truth conditions
if we do you know electrically stimulate
um part of your somatosensory cortex and
then you
your conscious self model is that i feel
a sensation of somebody touching my hand

because you stimulated that part of
my somatosensory cortex so on
the like conscious self-model level
i'm making sense of what
my neurons are telling me and my neurons
if we're asking what is it like to be a
neuron um is making sense of the
electrical signals
that it's receiving by saying they must
have come from
this area of the body because usually
when these bonds are stimulated in that
way
that corresponds to a sensation from the
sensory environment from this part of
the body
and if we
think of it that way then maybe we could
say that set of neurons
is predicting or doing
doing some sort of modeling
of the world because it's using these
signals
as a stand-in for the world these
electrical signals as a stand-in for the
part of the body that's usually
associated with that
or we could be instrumentalists about it
and just say that we're using our math
to
to describe how the neurons are doing
interesting and great to introduce that
experimental result that when you do
you know give a little electricity the
person will identify
that as being experienced a certain way
so um in s
and then stephen

um okay so i just wanted to address a
few of um
of thomas's questions
and just to give a little bit of context
with these uh particular questions
coming from i think it was from mark's
uh discussion with the discussion remark
last week
and we were talking about uh the body
uh being modeling itself and then this
came up
what about self-modeling uh on the
conscious level
what would that be um and here
i think that there are two things um so
uh thomas asked
what is more i think i asked okay
doesn't matter so
um modeling um here is important because
how is it that the self would be modeled
is important because all of us
understand what is
modeling from a scientific perspective
right
we develop our tools and we know how
things are modeled
right depending on the target system i'm
going to use certain set of tools
uh that's going to be dictated by the
system that i'm targeting
okay now we know what that kind of
modeling is i don't know what
modeling is when we turn to the self
i have no idea what that looks like is
it asian inference what kind of tools
are we talking about
that um the nervous system or the body
or even myself and this is the point of

this question
or even myself on a conscious level and
by conscious level i
don't even like the expression conscious
level um
here i was more thinking of the agent
level so the agent that interacts with
the world
and has um representations and things
about the world
is capable of drawing mapping things
that's um that's a sort of a much more
sophisticated kind of
interaction with the world that entails
language concepts logics
uh skills and that's the agent level
right so this is like lifting up this
conversation
into the agent level and thinking about
okay how is it
that self model modeling would be
motivated would it be
motivated at all um and if it is
how is it that it happens because um on
an agent level and i think this is
this is a a a point that might elucidate
things for us
on an agent level is an agent
ever engaging with beijing inference
i haven't learned bayesian inference
until very late in life
so before i learned beijing inference i
wasn't able to model myself
i guess so that's the point so even on
an
agent level for me to interact with the
world by using representations because i
can

on an agent level i don't think anyone
has an issue with that not even radical
in activism
not even any other an activist i don't
think um
so on a conscious level on an agent
level where we can actually engage in
representations and use
logics and use our concepts are we uh
using uh these kind of uh modeling that
we know what it is in science
so that's one point about modeling
the other one is um i think was shannon
that mentioned this very nice example
um and i think it's useful for us to
look at uh which is
um shannon said i feel a sensation of
someone touching
right touching touching my heart my arm
or something
right um so in that particular case this
is the agent level that's why i think
this is a nice example
in that particular case i feel someone
touching my arm
do i go around and uh start wondering
about
whether i felt that touch or not because
that would be starting off
an inference kind of kind of thinking
which we can totally do
by using logics and deduction and
induction and abduction all of that we
can totally do that
uh because we are enculturated beings
now do i wonder about whether someone
did
really touch my arm whether do i really

feel pain
because if i do wonder then i engage in
inference kind of thinking but that also
does not come for free that i'm using
beijing inference
unless i have the training and i'm
applying that for that purpose
right so uh what i think is problematic
about that is that
once we start thinking about um whether
someone really touched my arm
in kind of like a wandering kind of way
it's because we are assuming that we
don't have a direct access
to that particular experience somehow
that particular experience is hidden
from us
so we need to infer whether it is really
the case that i'm in pain
for example right so that puts us in
another
yet level of experience which is like
the position of a spectator
the spectator does not have direct
access to whatever is going on
so it needs to infer what is the most um
uh uh the the best explanation for that
particular case
yeah i just wanted to add that
interesting thank you
stephen and then
yeah that that idea of how much you have
have access to
and that may relate to different types
of engagement but the the
the way do we have access to our
multi-sensory integration
so we have a sense that we are i'm here

now i'm
i feel myself sitting i can see i can
hear
i'm aware of stuff and then i'm waiting
that's
in different ways just in my awareness i
suppose
and then there's the question of like
what i actually choose to do
um which made me more deductive maybe
more based on cues
so it is is interesting once you get to
this conscious
level is how much does the conscious
level give us
a way to not have to follow
um ergodic kind of flows
and we can kind of break out from that
but
when we get down to lower levels it kind
of needs to be more
flow it needs to basically be more
biological
so to speak um i don't know if that
makes sense but
thanks steven blue
so i just wonder is the model of the
self
maybe the ego like are those two things
the same
and like does the self model necessarily
need to be
um like rooted in narratives uh and i
think we
that we kind of touched on this a little
bit um last week i think scott had
brought that up and then just to reflect
really on what shannon said and what he

ness
uh just was talking about um you know in
terms of like a
a pain experience right like this is
kind of in like the
the eastern traditions pain is a very
like objective
experience like yes i'm feeling pain
like i just burned my hand on a hot pot
of water or whatever so like pain is
like an objective experience
but the suffering that's associated with
that pain is a subjective experience and
that's that's part of like
your ego or identity like oh my god i
feel so horrible for myself
or like suffering in relation to the
pain that you're experiencing is
optional right it's not uh it's not
something that you have to
necessarily like experience and so i
wonder then
is suffering like inference is is that
part that's the inference part or
or i don't know just a thought
yeah it's like the observation it's
raining and if you are i'm the kind of
person who's sad on a rainy day or i'm
the kind of person who's happy on a
rainy day
it's quite disjoint from it raining or
not but then sometimes when it's
biological and it's about our body
it's easy to conflate those two like
well pain is bad and it makes me suffer
because it's painful
but actually there is a little bit of a
step in between shannon

yeah thanks for that um so yeah i wonder
if there
it makes sense um to say to
a philosopher that we have direct access
to
something that something emergent
um but our brain is a spectator in this
process
so i have access to knowing that someone
touched my hands
i might question it if i see nobody's
touching my hand but someone stimulated
my brain in a certain way and i still
feel that sensation
i still have direct access i feel like
to that sensation
i don't feel like it's vertical with the
world anymore or vertical with
um touching on my actual hand
um and then maybe on the level of of
the person like blue said i can
associate
this feeling with pain or if this is an
instant where i enjoy this feeling then
i don't associate it with
um the suffering i get you said of pain
um
but my brain doesn't know that
like so my neurons are still being a
spectator on these sensations
and are still inferring
something about their causes or
something about what the next sensation
the next pattern of activity will be
given the pattern of activity they're
feeling now
so on our emergent on our side of these
emergent processes we can have

something like direct access but
within all of the individual parts that
make up that emergent process
we still have a bunch of spectators a
bunch of neurons
cool um thomas stephen and s
yeah so that's that's um that's very
interesting um
let's see i i initially had a just a
comment on
blue but now also on shannon so
um i think the first thing
to try and understand is um
what it really means to say that the
brain knows
anything at all um and that's actually
something in holi's work
that i would also quite strongly
criticize
because there is a weird way in which
a single organ in a body
suddenly becomes a an autonomous agent
where it is guessing things
where it is predicting things it's
knowing things and
doing all sorts of things that
we would usually relegate to the
personal level or the agent level um
this is i think fairly well known among
neuroscientists the
myriological fallacy ascribing
agent-level behaviors to a part of the
system
um the
thing is then that the brain doesn't
have access
and needs to predict or because it
doesn't know

[Music]
doesn't really make any sense anymore
when you
just see the brain as an organ in an
organism
and the organism as a whole has access
um so that's
that's one point the moment in what
you're doing here basically you're
shifting from
the uh from a neurocentric
way of doing cognitive science and
definition of the
mind to a more embodied one
um and then if you're talking about the
difference between
pain and suffering that comes with it as
blue was doing
um under the inactive approach
the um difference sort of falls away
um and both of them
are the very specific ways
that the sensory stimuli
and your own actions interrelate
those are called sensory motor
contingencies
um so if you get a pin prick
in your finger the pain
you're experiencing is the very specific
way
that the sensory stimuli in your finger
change as you poke the needle in
or the needle gets poked in if it's not
you doing it
or it's about squeezing a sponge and the
softness of the sponge that's a very
nice
example um you only really feel the

softness of the sponge
as you're squeezing the sponge there is
no
softness of the sponge separate
from your squeezing if you're not
squeezing the sponge
you won't know if it's soft and you
can't feel softness
so the softness you're experiencing just
is the particular way
that your sensory stimuli in your hand
interrelates with your own squeezing
so there's nothing over and above
that interactivity
or underneath that interactivity that
makes up the experience
interesting thanks steven in ass blue
thanks thomas that's really well put i
mean i think that's where
the idea of the spectator that in itself
is a
you can fall into that trap of it being
apart because like a spectator is
assuming like
i see the story coming at me and
therefore the brain is
having all this information come in
whereas if it's
part of the the mind is part of the body
and the proprioception and
and the dynamic with whatever is being
encountered
then that spectator role doesn't really
make as much sense so and i think that
um
that also then comes in with the active
inference idea that the signal isn't
coming in with the knowledge it's it's

being created
somehow well through active inference
through
this dynamical process so i think that's
really interesting and one thing that's
quite interesting to say with pain
um is is pain you can only create pain
you can only talk about pain through
metaphor it's a stabbing pain
it's an ache it's a it's um it's it's
kind of a
burning feeling so
all pain is expressed or has to be
expressed through some sort of metaphor
so
so um we have to utilize not just our
proprioception but we're
um we need to come back full loop with
our sensory apparatus to create the
thoughts
and this is where the mental space
psychology work could be quite
interesting to
to mix with that because that idea there
is that
there's an organizing principle in the
space around us
that we're effectively designing a
mental space
around us which then becomes
a part of that interface so anyway that
that's uh
definitely an important point
thanks ernest
um okay i wanted to uh just go
back to um brain level and then i want
to go back to the
self modeling um just to make a point

about that
um so on the on the case of the neural
level
um i think that we're talking about the
spectator and we're talking
we're trying to pursue this path where
we were motivating why the brain is a
spectator
and then thomas made up this nice
comment um
coming from um the reasons that jakob
hui provides
um to that direction and i just want to
address that from
um the free energy principle in the uh
the active inference um
um trend um so i think that even if we
would accept
that um we were under the assumption it
would accept the assumption
that the brain is a spectator and we
were sort of agreeing here with the
yakov huey
um that the brain is encapsulated in the
skull therefore it needs to infer
whatever is going on outside we have to
understand that the brain
has not uh gone to school to learn
asian inference or dynamical modeling so
that's important and
we also should not forget because this
is what brings us all here we're talking
about free energy principle active
inference and those things
so this is not something that active
inference or the free energy principle
will endorse
right that the brain is encapsulated um

so
uh according to the free energy
principle or active inference
um the brain is embedded in a
multi-scale system that
many people call can be captured by uh
by nested markov blankets
so this multi-scale system itself
organizes
at every single level and in
function of immediate influences
that we can explain for example by by by
by applying uh
active inference and and uh bayesian uh
and the
leaf and that kind of thing that we all
know so this means
uh one important thing which is that
every part of the brain
is interacting okay and
where there is a interaction there is no
spectator
there are these causal influences and
this is the principles of um
effective connectivity for example which
we talked about uh
about last last week so i just wanted to
put this out there that this sort of
like spectator encapsulation is
something that
would not be endorsed um and i think
that this is the right way to think
i'm just not sure if we should take the
realist um
view about it the other point that i
wanted to make just
going back to the self-modeling and then
probably just drop it but i think that

this is important for us think and it
wasn't mentioned
um the self-modeling it's important to
ask
who is doing the modeling in
self-modeling
who is doing the modeling is it a
homunculus
so i think this is important because
this there have been a few arguments
against this idea of like of
self-modeling
and um it's important to know always
because in science we know who's doing
the modeling is the scientist
in self-modeling i'm not sure who's
doing the modeling
yeah and yeah these are my points
thanks a lot blue and then i'll ask a
question from the live chat
so um i don't know the the brain as a
spectator
is is an interesting uh point and also
like who
is doing the modeling like these two
maybe might
go together but maybe are are
encompassed by a bigger question
um like what does the brain know versus
like what does the mind know and are
these two things
interchangeable like and what does the
free energy principle apply to does it
apply only to the brain
or can we scale it to maybe include the
mind
right or maybe the brain is part of the
mind if you take a more

embodied approach or what is the mind
has also been like positive to be maybe
like the bioelectric field that's been
out there and some work so
so what is the brain what is the mind
can it
is the brain part of the mind i would
say a pretty large part if you're
looking even at the bioelectric field
it's it's a big part right like so the
brain has most bioelectric
activity so maybe it scales to that um
so so and maybe like the integration of
all of the systems happens
outside of the brain i mean maybe a
component happens inside the brain but
maybe
then it scales up to include like
embodied knowledge such as you know
the gut brain axis or other um
areas so i just was wondering really
what the free energy principle as
developed by first and i guess is that a
brain specific
or do you what do you guys think can it
scale to include the mind or
or are these the same thing i don't know
just curious
yep um i'm gonna just wait for the live
chat because it looks like a lot of
people have thoughts on blue's question
so
stephen and s thomas and then i'll ask a
question
i mean this is also where the free
energy principle and the active
inference side comes in so i suppose
there's the active inference would be

like you could say it has to be inactive
so it's the whole mind-body environment
dynamic but then the question with the
brain is
is there a level of the brain
they should say it's all using free
energy principle but maybe there's a
level of the brain
which is um you know using free energy
principle approaches to create the
consciousness
as this kind of epi phenomenon that's
sitting on top of the whole organism
which is more broadly used in active
inference with this with the
with the um senses my feeling
is everything has to go back to the
mind-body environment
at some level but there may be some
i don't know some like slightly higher
level encapsulated piece that gives you
the ability to have this reflective
capacity of consciousness at like the
human level
but i'm not sure thanks steven
um ines and thomas
um yeah okay so i just wanted to um
address the mind brain um whether the
energy principle applies only to
mind or brain or both or only that
um i just wanted to bring to the
discussion
um the the the fact that the free energy
principle
uh is a principle that is supposed to
apply to every single system that is in
non-equilibrium steady states
and this is the majority of the world

it's very hard for us to find
a system a natural system that is not um
that is in equilibrium
um so that's what the free energy
applies to another way to say it is that
free energy applies to anything that has
a markov blanket in the sense that is
coupled with the environment exchanging
uh energy with the environment and
interacting with the environment in that
particular sense obviously these
things become much more complex to
explain for us when
uh we come to adaptive systems because
levels of complexity
uh sort of explode because of the the
the higher levels of degrees of freedom
right
so just to put it out there it applies
to um every single system that is in
non-equilibrium which is the majority of
the things
and then we can pursue a sort of a line
of research that
is about the mind brain identity
where we ask questions of whether it
applies only to the mind also applies to
the brain or
applies to both of them or tries to
unify both of them in a single
explanation right
um so i think that um
the machinery of the free energy
principle with the corollary after
active inference
um is is very nice to to to help us
understand both the mind and the brain
and here by mind i'm thinking about the

agent level
where we we can think about decision
making
beliefs uh what is motivating my actions
in terms of
more intellectual aspects of ways that i
can interact with the environment like
having thoughts and beliefs
uh that would be the mind and that would
be the nice way where i think that the
free energy principle with active
inference can be very helpful
um to develop um generative models for
us to understand
the specific psychological phenomenon
that is
on the agent level and then we can also
apply this
uh this machinery these generative
models to uh the brain level which would
be the neural level
where we will look at the connectivity
in
brain activity and obviously we can
explain these things by saying that
the activity in the brain can be
described
as if there was belief update as we know
uh uh
with uh with active inference what i
would then
be a little bit more careful about is to
say that
um this is precisely our neuronal
activity
or even the agent level uh works
uh when doing its own uh reasoning
um to act in the environment um so

that's what i would be a little bit more
careful about
is then the second the second step and i
don't think that we lose
a lot um i don't think that uh we lose
the explanatory power
as long as we understand that this is a
tool
thanks a lot thomas
yeah i was um basically going to
um say sort of the same thing as ines
did
um i mean that makes it a little easier
for me
um i think just mostly what is also what
i would like to
add is the free energy principle isn't
going to tell you
where the mind starts and where it ends
um since it does apply to just about
anything that exists um
you can just take any system draw your
own boundaries and start analyzing and
things will start popping up
um when you want to think about where is
the mind is it the body
is it the body in an environment is it
just the brain
um those are philosophical questions
that will need philosophical answers
um
you can answer that in in different ways
and if you want to say it's just the
brain you're going to
end up in sort of always camp the
predictive mind you want to say it's the
the body in
its environment um you're going to

end up more in an inactive camp but
those are
separate questions that i i don't think
you're going to answer just by looking
at the formalism
cool it's like should the title of your
book be the predictive mind
the predictive brain the predictive
brain body mind the predictive niche
you could keep going and the point is
when we're taking this instrumentalist
multi-scale nested markov blanket
approach you just
could kind of pick your title depending
on what you want to emphasize so the
question
from the chat which i think was related
to this question about
um who is doing the self-modeling and
who am i which
sounded a lot like some meditation
prompts the question
from cv in the chat was can
spontaneously arising thought give us
insight in to answer the questions in
this discussion that cannot be given by
any other research or other thought
types
if yes where and how thanks a lot
so as selfs if we're thinking about this
self modeling at the conscious level
thomas you just said maybe there's
things the formalism isn't going to
resolve
for us there's uncertainties we have
that aren't going to be
equations resolving it for us um
might spontaneously arising thought

or other types of reflection maybe
shine some light into any of these
questions
so i'll just put that out there and go
to just leave it up
while we go to another um slide and then
stephen just
just to put this up because this was
some related questions about the body's
expectations and about where these
priors or these expectations come from
so any thoughts on the spontaneously
arising thought
or any other topic stephen and then any
other hands
i think this is quite interesting from
the point of view of how
neurophenomenology is generally done
with people doing meditation
so there's the idea that you know you
track your thoughts that come up
and then you sort of capture them but
the question is well what's going on
when i'm in the flow
what's going on in my pre-reflective
state when i'm in action
and interaction a bit like now where in
many ways that's what most of my life is
and most people's life is and actually
where maybe a lot of the interest is
because that's where we struggle
or have opportunities so i think this um
question about how also is
how are we when we're in this other sort
of dynamic
and how can we go back and revisit or
explore
what it is to be in action and what's

going on there
so i think that that this these same
questions sort of feed back into that
um related to the flow state and what it
means to be a self
are we most in touch with ourselves are
we accessing ourselves being ourself
most when we're
writing a personal statement or when
we're really focused on you know
downhill skiing or a chess game
the kinds of things that suzette mahali
wrote about in flow
and so during that downhill skiing or
the chess game
maybe somebody is not thinking about
some of their somatosensory or their
narrative self
there's aspects of self that are
actually specifically turned down to
like almost zero
and then there's a different focus but
then maybe somebody might self-report
after the skiing i'm so happy that's
when i feel
like myself and so what is this self
that we're getting in touch with and how
is that related
to our um uh spontaneously arising
thoughts blue and then anyone else
so just to kind of comment on that and
then tie it back into
um like like what who what is the self
that's doing the modeling right
so the flow state is super interesting
and
it's like that is the space where the
ego

kind of drops away right like when
you're in downhill skiing like you're
not thinking about like
what my colleagues thought about what i
said at the business meeting
like it's totally not at all in your in
your like mental
it's like all you're doing is focused on
that task and especially in kinesthetic
um like situations like dance and
drumming
and and all of these things which also
tend to like alleviate the trauma
experience right like so all these like
where the flow state happens the like
that's where there's really not that ego
so there's not that
um like self doing the modeling but then
the ego might go back and be like i'm an
awesome downhill skier
right like so so there's this right like
self-reflection
on the flow state that that is like
is that the integrator right so it just
kind of maybe
gives more um like rise to the ego
doing the narrative or the modeling when
you're doing that self
self modeling or or self narration right
yep really interesting i'll throw
another
question from marco in the chat
marco wrote how about taking mutual
inference more seriously and posit that
mutualistic interactions
are legitimate perceptual entities
which may possess the potential to
realize a markov blanket in turn

what is the maximum markov blanket like
i noted last session
i don't think that the x or so kind of
the logical boolean gate
attitude to the schema and categories of
realist instrumentalists
um uh basically uh but
i just can't read it live but yeah like
it's should we be looking for a
disjointness
between realism and instrumentalism um
that's kind of the topic of this paper
so
how do we take a realist approach to the
instrumentalists
um is almost what marco is
uh getting out there but anyone can
take a thought on that um thomas and
then in s
um i was mostly going to ask where is
the chat
where could i perhaps read this myself
uh to try and sort out what exactly is
uh
oh it's the youtube live chat so if you
just go to our
twitter look at the most recent post
you'll find the link to there but i'm
not going to put it in the jitsi
chat here because it will make like a
swoosh sound but
just go to the youtube link from our
twitter um yeah
thank you yes
i just want to uh sort of like address
that by
sort of um throwing a question
at it or sort of a comment actually um

i want to say that again in that regard
i want to just point out that we
the neuroscientists are the only ones
who have the cake
and want to eat the cake and that means
that
amongst every scientist that is working
with models
across the world in every single science
we neuroscientists are the only ones
that are realists
very nice and i'm gonna take that to
this next question so
as i understand that it's almost like
there's the venn diagram of people who
are
studying models or using models and
basically no one has a problem with oh
yeah
the flow of water through this dam is
going to be modeled with this kind of an
oscillator
everyone's pretty instrumental about and
then you have people who are thinking
about
the brain and there it's kind of like
the stakes are high
because it's it's us on the table um
dissecting
ourself and so it's hard to step away
from a realist interpretation from
what's
you know right inside so how do we step
outside of that and then this is where i
um put with uh alex's uh suggestions and
prodding about where engineering fits
into this and almost
is engineering like radical

instrumentalism
when we're dispensing with any kind of
realism and we just want to get it done
within
the time frame and the budget and the
affordances that we have
is that akin to engineering so blue
steven and then anyone else
so i i love that neuroscientists
have their cake and want to eat it also
and i just was like thinking and
reflecting like maybe there are actually
some other groups where that applies
like
medical doctors um also like
kinesthesiologists right so anybody that
kind of studies like the brain
or the body like does it apply like they
have the model they're also using the
model like maybe they're not
modeling the ma or modeling the model
like within the model i don't i don't
know like it's like it gets like very
nested with neuroscience but
but there maybe are some other um groups
that are doing this as well
great stephen i would
i would actually question that because
i'm having some challenges around
working with complex adaptive system
applications in other fields
you know like social development and
stuff
and i think that this is the they may be
the only ones that
try and get the cake and eat it but
everyone is just swimming in cake mix
in a funny way because everyone assumes

you can make a system model
and that the system model can be a
snapshot
and you have flows of real things
flowing it's very entity based
and even the idea of the self
and um the the idea of the self
and there's a good paper on this by
bolus
and where he talks about the self is
actually a social construct in itself
about two and a half thousand years ago
the idea of self
was constructed and i have a sense of
self
so i'm in a society and if you go back
to indigenous approaches
i you are one of the land you and the
land are one
which is maybe a bit more like reality
if we're gonna
say reality but there is this challenge
because
everyone's using systems at some levels
to make the models or we talk about that
way so i
kind of agree but i would also say that
it's the the there is a paradigm
challenge here about how to think about
um what's going on in terms of how
we are actually knowing about the world
thanks steven um blue and then anyone
else
so that just reminded me like the self
as a social construct
um it's a super interesting point and i
don't know if you guys are familiar with
the work of

jay garfield but he talks a lot about
the work
of um like or the development of the
self like
like becoming self aware or self
consciousness
like in terms of like a developmental
aspect and like you don't realize
like like the context that you realize
that you are
a self like that's distinct it's
actually not
yourself that you realize first you
actually realize
others so he talks about second person
so like you
as soon as you realize as an infant that
there's a you who are going to come take
care of me
like there's a you out there that are
going to feed me there's the you out
there that are going to
make me comfortable when i'm you know
sick or wet or whatever like
in in very infant hood in the human
species
as soon as we realize that there's a you
that's when
the dichotomy appears that there's me
right like so it's just very interesting
because it
it starts as a social construct this
idea of the self from a very very
early age thanks blue
and alex v yeah thanks
i want to add from engineering domain
that
for now the system engineering have a

sad joke that we can we can't include
people
to the scope of the system because we
can't control people
so okay we can control almost everything
but not people
in the loop so we are trying to
go out of the way with it
but but if
engineering is a radical instrumentalism
we can consider active inference
framework
which provides as it was said on
agent level some models
which provide this opportunity to
include people
to the scope of the system uh because
and by organizing some communication
structure
around this agent and create some niche
for them where we can
communicate in uh some
language of the domain we are working on
and here comes
to replace all stuff
with languages and with ontologies
and also for ontologies and knowledge
life cycle management uh and it's
if we consider it in engineering way and
not trying to go uh
down from agent level uh but trying to
see
or on agent environment and maybe it's
as it was told that we can say
about agent level how mind level
and for this mind level uh the behavior
of the agent is a correlate for his all
kind of states under his

marco blanket thanks
alex it's almost like because you can't
engineer the human
you're engineering the inputs and the
affordances that's where active
inference comes into play it's like if
you're going to do water
engineering you're talking about making
the canal and the dams and the
measurement devices
not going in there and engineering the
water water interactions
so engineering and instrumentalism is
almost always
at the periphery of an uncontrollable
system interfacing between
our regimes of order and attention and
then a system
whose dynamics are being modeled
instrumentally because
it's um the fringe it's what's not known
so um ines and then dave
yeah i think that's um that's about
right because so we have a tendency to
think that
to take the realist approach if we come
from the sciences
of the mind and brain but then we would
not commit the same kind of
um would not have be have the same kind
of attitude
if we were trying to model uh
a non-adaptive um system
so i think that comes from the fact that
we do see that there's a difference
between adaptive and non-adaptive
systems
and that sort of like gives us um

part of the motivation to think that
because they're adaptive
then they must be uh using the models
that we
are in order to explain them but i think
this is a fallacy that sort of conflates
um uh the the the explanation
exponential
so um but the point that i wanted to
make uh was more of
um coming back
to the self as a social construct
we can push that even a little bit
further if we go back to matarana and
varela with the autopay as this theory
and i find this very
useful uh because it's it's it's a nice
exercise all of us take um cognition
to be something that is an ontological
property of
uh the organism and this is precisely
what matarana specifically
uh challenges with the autopias's theory
um
it tells us that cognition is a
construct
that is in the eye of the scientists so
cognition is not even something that has
a reference
um in the world which is super
interesting because
when we started constructing things like
this that this is going to have direct
implications for the way we see models
uh what it tells us is that we
observe behavior in
uh in systems as they interact because
they are adaptive systems that interact

with the environment and we observe
certain behavior and
make sense of it by by by coming up with
patterns
um that's how we make sense of this
behavior we come up with patterns and
then
we deem these patterns or this behavior
as
cognitive but we do this as scientists
the
system itself the organism itself is not
cognitive this is not an ontological
property
of of the system so i think this is this
is important to think about
because this also allows us to now by
taking the kind of same reasoning
then we can apply this by saying that we
can model we can observe
this this behavior we can patent this
behavior
through the modeling techniques that we
have but the modeling technique is not
an ontological property
of the organism
thanks for that dave and then anyone
else
if you want a real grasp of where
a self comes from you got to start with
the newborn alum shore did some
very detailed work on the creation of
the self
as part of the mother newborn dyad
and the really intricate um
not self modeling but us modeling
and a constant dialectic
of um

alienation followed by reparation you
know
mommy dis disappoints the baby and it's
the worst thing in the world the baby
wants to kill mommy and then wants to
eat
mommy and wants to save mommy from the
monsters
well once you got a good grasp of that
then you go and do uh
ethological work and see how other
creatures
differentiate their children there's a
lot of socialization a lot of
creation of of the style of a newborn
cult even though they can get up and run
around pretty fast and then you come
back and look at pathological cases with
humans melanie klein was the first
person who realized that you can
psychoanalyze a child who cannot speak
yet
and really worked at that and figured
out how you can find out what's going on
in the kid who's the
who's the the baby mad at by seeing how
how he plays
with uh toys how he plays with dolls
um his pictures these kind of kids that
can't
uh speak can still draw and
you know rip up pictures that they don't
like and um
there's a lot of poignance
in the emergence of the self-creation of
a self
cool well um to kind of operationalize
that and bring it back to a figure and a

question in the chat
this nested markov blanket uh
self concept or in the neighborhood of
the self
someone writes in the chat can we
measure markov blankets
the way we measure or estimate any other
physical entity
i've heard carl fristen saying that
every electron does have a markov
blanket
can we measure statistically speaking
electrons markov blanket
and i i think it might be related to
some other points that marco has in the
chat which we can bring up afterwards
but
it is really important if we're talking
about markov blankets
um and that's figure one is this this
pink dashed line
that um is uh and if that's how we're
going to be
modeling in the free energy principle
how do we do inference on these markov
blankets
what does have this and um how do we go
from there
so um shannon and s
yeah so this is a good question we
we we can with a caveat measure the
markov blanket of
of a certain electron or certain neuron
it's
the s statistical boundary here
and it's any states that you need to
describe
to in order in order to explain the

behavior of what you want to model
and you can kind of decide as a
scientist at what level
do you want to describe this behavior so
your electrons markov blanket might not
include the behavior of the body it
might just
include the neighboring electrons next
to it or
as far basically as far out as you need
to explain the behavior
at the level that you want to explain it
so it's not a fact of
of nature which is me agreeing like
wholeheartedly with this instrumentalist
perspective
it's where you want to draw the boundary
in order to explain the behavior that
you're interested in
thanks shannon and us
yeah that's precisely right um i wanna
just to say that
uh marco blankets are not something that
exists out there in the wild
um so when we say that we can measure
the markov blanket of a neuron it's i
would probably just like
formulate it differently we can apply um
markov blanket as a statistical tool
uh to the level of description of neuron
which is the one i'm interested
and it is precisely this special tool
this statistical tool which
does not have to correspond to physical
boundaries of anything
because um we get markov blankets by
identifying the influences that certain
states have

in other states so that's how we find
markov blankets they don't exist they
are a way of us
to um sort of like have a nice topology
of how certain states are influencing
other states and in the end we got would
also get like a nice thing which is like
how a certain system is coupled with
another system and
because of the it's skill-free um as uh
as
shannon was uh rightly pointing out
because of the scale-free property of
the markov blankets
we can apply this statistical tool to
whichever level of description we are
interested in
studying so we can apply this to the
single-level
neuron but we can also apply this to
a column we can apply this to regions
can apply to the brain as a whole we can
apply it to the organism
that is uh coupled with the environment
and we can also apply this to
social networks and things like that as
long as there are interactions
and influences um uh that are occurring
between
states then we can um we can apply this
tool
to explain how uh these influences are
occurring
um between the states and yes this would
be a very instrumentalist reading
and i think the only possible one
cool and and marco um adds a few more
points

and says i don't buy the exclusionist
categorization
when it comes to fep because it
functions both as a modeling
technique and an ontological system and
i think
we're um hopefully conveying what you're
writing and it's a discussion that is
non
it's a yes and with realism and
instrumentalism and so
um i hope we're bringing a lot of
perspectives to bear
on maybe where the fdp sits in this
space and that's some of the
great topics that were broached in the
paper and some of the really important
work that's being done so that we don't
preemptively come
hard on one side or have an exclusionist
perspective and
part of that is actually enacting that
conversation
not just having one really convincing
thought that convinces us that it is
way a or way b or neither a or b but
actually bringing together
different perspectives on the topic and
then seeing where that goes so thanks
everyone
you know as always for participating
which is actually
enacts the pluralism through
participation
so stephen and then thomas
we're kind of one good thing about this
paper which
is opening up even bigger aspects of

complex adaptive systems
applications so i feel like we're we're
putting a lot of
bigger ideas on the edge of some other
fields of practice
on this work that neuroscientist is
doing but i think it's really valuable
because
there's this question of
non-representational approaches
um in social sciences which are kind of
in that
ways of trying to understand things
around complex
complexity has been used more and one of
the big areas
by michael quinn patton in qualitative
research is
there's this new work around principles
focused evaluation
which moves away from more sort of
betterment or sort of final product
evaluation this idea of looking at broad
principles
for projects which are evolving but
haven't got a set of rules that can just
be governed and replicated it's
it's it's formed you know what it is
it's not completely
being formed from nothing but then
you've got to work with the principles
and i think that may
reflect some of this um discussion
that's going on here with with
rules and principles because i'm now
thinking well rules are probably
something that's more realist
and principles maybe

are broader than that i don't know
i think there's something around this
broader area of complex adaptive systems
in general and what it means to look at
them and then this is one aspect of that
awesome stephen thanks a lot thomas and
ines
yeah um so i wanted to quickly add on to
what um shannon and enes were saying um
in response the measurement question
and in full agreement just to
really answer the question can you
measure a markov blanket in the same way
that we can
measure any other physical system that
would then be a
no because the markov blanket is part of
our toolkit
with which we measure the system
um and then
on the marcos
commented on the exclusionary reading of
uh realism and instrumentalism and uh
either to pick one or the other
um
that's a very interesting point
and i guess shows that the
enumeration we had of different
positions um
wasn't exhaustive i had not personally
really considered a pluralist account
and i'm not sure exactly what that means
still
which is i guess part of why i didn't
consider it um and i think i think that
will really take some
some um good spelling out of what
pluralism really means here because

the only way i can think of it is
different strands of realism
because i don't feel like you can get
any form of realism if you actually take
instrumentalism seriously
in the sense that if you're saying well
this is part of our scientific toolkit
with which we're describing a system
realism is cut off right then and there
[Applause]
but of course pluralism in the sense of
let's
discuss different ideas and look where
that leads us
is a different thing and we should
definitely do that because otherwise you
know we're not really
going anywhere yeah that was a yeah
very very interesting it's like
conversational pluralism or a social
pluralism
exploratory pluralism is a slightly
different kind of claim
than the um other
sub questions that pluralism could
impinge upon
an s and then anyone else who raised
their hand
okay so i wanted to address the question
in the chat
about whether we should be realist about
free energy principle
um and yeah i want to just start by
saying that the free energy principle is
a principle
so we would expect it to have like
something to say about
uh a tell us of things some explanation

that we would expect
systems to behave in such a way and this
is what we discussed last week
there was a distinction between the
furniture principle and process theories
right process theories aim to describe
processes
uh that would explain a certain uh
psychological or cognitive or neural
phenomenon or something like that
the free energy principle has something
to say teleologically speaking
so it has something to say about how we
would expect
an organism to behave towards something
some ultimate goal
right and this is for example uh that
the system
does not want to die wants to avoid
uncertain states uh
because it doesn't want to die so he
wants to select the states that are
going to be much
much more favorable to um to their
their existence uh in a particular
situation so obviously the free energy
principle
has got these sort of like much more um
is charged with
with giving us an explanation about uh
whichever we find the system doing the
system is going to be
um very likely is going to be evolving
towards
uh avoiding its dissipation so that's
what we get from as opposed to
the the process theories where we don't
really get to tell us

what we get is um a description of
system of processes uh in systems
that minimize prediction error but let's
think about this and let's see what
would be a realist
uh take on the free energy principle
right
um so the furniture principle is about
systems or adaptive systems it can also
apply as we said to all natural systems
but
let's take the adaptive systems here
because they're much more complex and
that's what we are interested
on um so the free energy principle is
about systems that
minimize free energy expectation right
precisely because they want to do that
that's where we find the telos
um so then i want to just to
draw our attention to the fact that
these explanation these tell us
applies to all adaptive systems
and that means that it applies to
bacteria
um and uh there's been also a paper with
pristine and calvo
that um it also applies to plants
and um it also applies to systems with
with nervous systems
right so a realist account of the
furniture principle would say that
uh every single adaptive system is
minimizing free energy that's a realist
account of the free energy principle
now the question that follows is the one
that relates to the paper that
uh thomas and i wrote is okay so then

um our representation is going to be
entailed
in what level the representations need
nervous systems
because one because this is the free
energy principle applies to
all adaptive systems not only to the
systems that have nervous systems
so if we take the realist path about the
free energy principle as something that
uh systems really do they they minimize
they attempt to minimize this free
energy
um then this applies to all
uh adaptive systems so where does that
leave us
to the representation list account
very thought provoking so maybe another
way to say
if i'm hearing it is like if it is the
case
that the brain for example is minimizing
free energy so that's the realist
take on the free energy principle then
there have to be representations in the
brain
that correspond to
free energy as being some sort of
quantity that's being
carried in the brain whereas to do
instrumentalist free energy principle
one doesn't need to have any specific
um stance on whether representations of
any kind exist
because it's being clarified that it's
just like well the linear regression
exists on my calculator not in the brain
so

yeah is it is that
i hope really okay stephen and then
anyone else and also yeah
great points happening in the chat
there's a lot happening about
um for example natural kinds are they
disjoint or are they
isomorphic with marcon blankets there's
a lot of great points that people read
and consider
there stephen and anyone else i think
one of the advantages the free energy
principle has and it's less clear once
you start getting into the process
theories is if
if through this dynamical interf
interplay between external states
internal states
and markov blanket states if it's
working with
entropy entropy can be no dimensional
so you've got this advantage that it
could be happening
but and you could be inferring
information by
somehow extracting
in whatever form somehow
knowledge through a change in the
randomness
so it's by a change in the randomness
that you get a signal there isn't
actually a signal
you may you may not have that problem
with representationalism
coming up even if free energy principle
is actually
kind of happening because it's not clear
how much it's become something with

dimensions
so i think that the question then is
once you
go beyond that into the process theory
maybe
um yeah some parts of that as well are
kind of dimensionless there's still that
flow
the nestedness that the nature of things
traversed in between
and then there is a sense that once you
create structures that you do have
physical structures and you get this
realist and you have to make a choice
so maybe i don't know that's just my
thought on that
and um it's almost like is the heart of
the matter
how something is or how we should act
because if
the heart of the matter is framed as
being i won't be satisfied till i get
the realist explanation i won't be
satisfied with what an electron
is until the heart of the matter has
been reached whether it
is a particle whether it is a field and
then if the heart of the matter
were actually framed in an
action-oriented way
so the heart of the matter is electrons
you could go way down the rabbit hole
and sub-electronic stuff
but the heart of the matter is how we
engage with electrons the heart of the
matter with a person is how we engage
with them
in that case it might be possible to

have adequate
if not excellent policy with a very
partial instrumentalist reading
you could say just with the models that
we had in the 1980s
we had instrumentalist models of the
electron that allowed us to build
transistors of this power that were
required for this task
so just taking the engineering take and
so again by kind of making the heart of
the issue about how to act rather than
essence and identity
some of these questions that are related
to representationalism essentialism
internalism reductionism all of that is
aligned with
the heart of the issue being those
topics and so it's always a
countervailing argument to take a
relational
or an interactive stance against that
broader
kind of philosophical gestalt versus the
action-oriented turn the pragmatic turn
engineering as a philosophy or as a
as a way of life and then thinking about
organisms as more like
engineers in their niche like ecosystem
engineers
niche modification evolution is
selecting for good engineers
not the good philosopher not the bird
who thinks
about how to classify the seasons but
the bird who reacts
to the um niche in a way that leads to
continued success

so it's just very interesting like
almost a question arising is we talked
about these axes of variation
and this paper in figure one or table
one laid out the two by two
and so we can kind of lay them out
orthogonally here but there's a lot of
these dimensions and some combinations
two-way combinations or three-way
combinations might be
um very elucidating considering well no
one has thought about this combination
of yes
yes yes no or which of these axes are
kind of tied up axiomatically
that's pretty cool to think about if
anyone raises their hand
we can um take it a different way i'll
just go through the final
little slides that we had so one slide
says where does design fit into all of
this
is design instrumental or realist and
what about technology so we kind of
talked about engineering
but what about design and all of its
meanings and then
another branch we had explored a bit was
we kind of talked about it earlier today
as well is
how can we really distill some of these
uh claims and so that that could lay it
out on a
sort of shared table where we could come
together with all of these different
philosophical perspectives and
experimentalist perspectives
how could we reduce this to a set of

claims where someone says actually
um i disagree with claim two or i would
like to add claim three
and so that style of getting through
these topics would be pretty
cool so with about you know any amount
of time that we want to have
left what's something that somebody
would like to
pick up on or head off in direction of
in s and then anyone else thanks
yeah i can try to unpack a little bit uh
claim number two
so i think that here when we say that
humans do
inference i think that's precisely right
but we can unpack this a little bit
further
we can say that humans are capable in uh
everyday life because they are in
culturated systems they're able to
engage
in inference uh reasoning these logics
and concepts etc that we already uh
mentioned right so that's what we can do
but uh humans are also scientists and
they also receive training
um in modeling in general so to speak
and once they do uh scientists
uh can also do other kinds of inference
which entail
much more sophisticated kind of
inference such as all these tools that
we've been talking about like
traditional bays
and and and that kind of thing so i
think that
it's important to say that in everyday

life we can engage in inference that
kind of
inference uh does entail um
representational properties because we
are trying to build up a representation
on the agent level about something that
we puzzled about
and we can also do uh engaging this kind
of inference as
scientists with much more sophisticated
kind of tools
cool um so
anyone else have any
thoughts on that on this slide or let's
just kind of give a quick another
uh coat of paint on some of these topics
because
a lot has been raised in this
paper um i'd like to return to the
roadmap
and just think about what this paper did
and ask the authors what their next
ideas
just to any level of detail or however
you want to convey it
what are your next exciting questions in
this area
or how did finishing this paper
prepare you for the next lines of
research or investigation or
collaboration that you're going to be
involved in
either either thomas or ness or anyone
else how is it going to change their
research after reading this paper
actually
um i'll go i'll go first then um
well hopefully i'll soon be finishing up

my phd
and even more hopefully starting a
postdoc to actually be able to continue
all of this
uh properly um but i'm now working on
a paper on anticipation
um and in part i am trying to
put the free energy principle to use and
i'm having a look into how the
notion of an information geometry can
help us
explain formalize or
describe
the sorts of statistical patterns
that agents become sensitive to when
they
anticipate
so that's one of the lines that i'm
hoping to make use of the
free energy principle in an
instrumentalist sense
cool thanks for phrasing it within
the uh framing of the paper i guess
ines and then we'll kind of move into
the non-author
collaborator range go ahead dennis
oh sorry you're muted
here we go you can't hear me now yeah
yep okay so one exciting thing that
thomas and i
are working on and as a follow-up uh
together with uh with a with another
colleague
um is precisely on
how to move forward in cognition um what
would
um this instrument uh
would be put to a good use in

understanding cognition uh as taking
taking models to be
in a very instrumentalist kind of way so
what we want to say now is uh that
moving forward
then models of cognition should not
um attempt to uh capture or represent
the properties of cognition of what
cognition is
but instead the very idiosyncratic
patterns of activity
in a particular system um so what we are
challenging is the representational
models of cognition
that we don't think that um makes sense
um
because for one um cognition sorry
for one cognitive scientific work is not
a perspective from nowhere
and for two um cognition is not
a fixed structure in a closed system so
we take cognition
in a very kind of dynamical system
stochastic system kind of way
so then we are trying to now develop the
sort of like
models that we think would be the most
appropriate models to capture these
aspects of cognition
sounds really interesting i look forward
to hearing about how you keep the
general perspective while also diving
into the minute
particulars stephen and then anyone else
yeah this is i think it's helpful to
hear this
thinking about modeling because trying
to bring modelling into other contexts

so i'd be interested like
when there was when thomas there was
talking about
using information geometry and to think
about how to model
systems and i think ines is saying the
same thing is it that you're
not thinking about modeling the system
so you can describe
the realism of the brain but it's like
how can you get the dynamics so it could
be useful
in i don't know computational psychiatry
or
it could when you say system you're
trying to see how the
dynamics reveal
not necessarily a realist interpretation
of the brain
in the bigger system but just something
useful about the bigger system is it
kind of like
being more applied in some way
to applications in in the world
or is that i'll be curious on your
thoughts on that
yes um yeah okay so yeah so it's
basically following exactly the same
reasoning
which is um coming from the view
that we've come uh which is that
cognition
is um observed in in behavior in
cognitive behavior right and we try to
make sense of that
but cognition is not something that is
uh static is something that is dynamic
and uh evolves in into a certain way

because systems uh
adapt because systems have this
particular feature
so um what we think
it's important to look at is uh
precisely how can we explain these
dynamics
of this cognitive behavior instead
of from having models that aim to
capture
the properties of cognition in a very
generalized kind of way
but much more specified into how can we
explain this particular map this
particular um
cognitive activity and these can be
applied
in in cognitive psychology labs but it
can also apply
this particular way of thinking where
models
should capture these dynamics and this
is very much the dynamical system
paradigm
uh where models uh uh try to capture the
dynamics
um obviously this entails uh um
a hypothesis and a theory about what we
think
is the best explanation for that
particular behavior that we observed and
this could be like in
again cognitive psychology labs in in in
perceptual learning paradigms for
example
but it can also be as we spoke about uh
last week this can also apply to
um brain data analysis because that's

precisely what
we don't know is how the brain generates
that kind of data
and then we can take different positions
we can look at
uh mapping the structures which is a
very kind of like static
kind of thing to do because we want a
topology nice apology of like the brain
as a map
which is also very useful to have but is
very descriptive
so basically here we are dealing with
description versus
explanation because once you have a
model that is a hypothesis driven
then you have a theory that you want to
test and that's what gets us
into um all the variational base kind of
area and where you need those kind of
tools because we are working with a lot
of
theories and assumptions and each model
is precisely going to be
um a theory about what we think that
perhaps has generated the data which is
what we actually have
access to so that's the kind of
yeah overview
here's one thought um anyone else can
raise their hand on actually where this
philosophy to clinical applications
comes into effect so let's just say that
there's some debate like about
molecular reductionism that's unresolved
in philosophy land
and then that percolates into the
clinics before the philosophy is even

clarified
like about genetic causation so you have
genetic counselors who are doing genetic
medicine
while there still debates what is a gene
and what is causation should we think
about genetic reductionism or holism
that uncertainty is actually in the
research
paradigm in the comparability of results
in the basic research is going to lead
to
an uh inaccuracy in communication and
in confidence at the clinical level in
the genetic case
now with as fep we're kind of talking
about neuroimaging in a big way
also other antaroceptive modalities are
being expanded
but really a lot of the foundation of
this is in neuroimaging and neuroscience
and so if there is a mess
philosophically then the research is
going to be less coherent the
communication is going to be less clear
it's going to lead to clinical
inaccuracy
where as if it were the case that there
was a clean philosophical way
at least to even frame the situation
perhaps that could lead to more
interoperable research
agendas driven by those who do and do
not wish to engage in the philosophy
specifically
and then increase clarity of like
recommendation or of communication
at the clinical level with people who

are multiple steps away
from the paper but we could actually
reference them this exact paper
for example if someone said we're
hypothesizing that people who have this
connectivity pattern in the dynamics of
the spm it's not about how your brain's
connected versus someone else it's
hypothesis about a dynamical system just
a thought
an s and then anyone else
yeah i actually just wanted to read out
loud uh
a quote by danielle dennett that i think
just
really captures what you daniel uh was
saying
not only in the clinical setting but
also in in
particular is referring to to science
and i think that
yeah i very much agree i think it's very
insightful he says
um in his book that uh let me see if i
oh goodness i lost it
um so you can take a second to find it
yes yeah
anyone else can raise their hand um in
the live chat people can provide any
final thoughts or questions but this is
just really an awesome
series i appreciate the authors for
joining us both sessions for all the
participants who engaged with this
paper and we're already seeing how this
debate like is modifying the way we
think about the topics so ines
oh yeah i've found the quote and

um well he tells us that um uh
the topic of representation should be um
oh no sorry it tells us there's no such
thing as uh philosophy-free science
there is only science whose
philosophical baggage is taken on board
without examination
and i think that really nicely captures
what you were saying daniel
nice thanks a lot for sharing that
shannon
i think um at the risk of bringing a new
topic with
less than 15 minutes um to talk but this
is
a big question so
i don't know in in neuroscience maybe
there's this just
pessimistic understanding that what the
philosophers and neuroscientists say
once it trickles down to the public it's
going to be
misconstrued so we're just like
constantly
reteaching the general public and if a
doctor is con
conversing with a patient they're
constantly saying
in this instance this is the thing
that's happening in our brain that
matters for this diagnosis that we have
to do to help you
solve problem or something um
but at a like a moral and societal
level in every aspect of science this
debate between whether we're looking at
a true
natural kind or just something that we

are taking a certain stance about as
scientists as a society
um has huge implications for policy i'm
thinking
of whether gender is real or not
and it has huge implications for policy
that like
that very detrimental like it hurts
people
if you take a certain stance that
there's a certain gene that codes
for a gender when
that concept itself is is a definition
that might not be a
complete scientific truth but some one
somewhere wants to say there's a
scientific truth
that gene means gender means policy
means
and then you just have this cycle um so
even though this seems like a very you
know
maybe we could just say ah i don't agree
with the philosopher or i don't agree
with neuroscience and leave the debate
um the fact that the debate exists
permeates into the rest of society and
culture and
is it's important maybe to say that
science
science maybe even as a whole isn't
realist about the world
science is instrumentalist about the
best way that we can interact
in our world and you do have to bring in
that baggage that daniel dennett was
discussing and this would
also include moral baggage

[Music]
thanks shannon and i posted this book in
the chat
studying human behavior how scientists
investigate aggression and sexuality
by professor helen longno so taking a
course with her in grad school and
reading this book
the way that it framed pluralism and
the sciences and the different
methodologies and just
laid bare in some ways how these issues
it's kind of like having the cake and
eating it too it's like these are the
issues that are
so near and dear to our social reality
and so it's hard to take the right
stances philosophically when it's easy
to
um lose sight of what's really happening
sometimes and just how things are
and like of course it's the y chromosome
it's like i mean it's an allusion to
genetic reductionism
but it's a little bit broader or it's
different than that and it's a yes and
with genetic and non and it just goes so
much beyond
a simple reductionist claim that it's
easy to get lost
so how can we even use active inference
and the communication channels of public
engagement
to actually at least within our sub
domains or
informational niches at first how can we
bridge that gap in a new way
it's an interesting question stephen and

then anyone else
yeah thanks for this excellent
conversation i suppose
this question about whether we can not
just take some realist
neural correlation as our way in
wasn't even an option before it was like
it's pretty it was basically that was
the only cake on the table
so people just took it and now it's like
well actually we can
we've got this other way of thinking you
know so
the rate the way we're doing it before
isn't the only way
it's just it was the only way so i think
that's one thing and
it's helping me because um
i'm seeing people in this area of mental
space psychology
trying to prove their field by saying
say peripersonal space neurons are
firing
and they're not necessary they're trying
to use this
these brain processes as evidence you
know and it's like if you don't show
causal you know and i mean i even see
that in my own
thing i think there's a tendency for me
in my work with spatial approaches
how can i explain this and it's like and
if you can't what you left with you're
just left with
well not a lot sometimes so this does
give
another way to make um use
of uh phenomenological

wisdom i think so that's what that's
what i'm taking away from this so thank
you
yeah thanks for connecting it to your
work and experience
blue and thomas
so i just wanted to thank enes for that
quote um
i thought that that was like really deep
and important and like a good
understanding really of
um the philosophy that's like behind
uh all of science might really enable
um us to go forward as a society like
i've just noticed with like covet and um
like at least in america and stuff
that's been going on in our country like
people are are upset that the science is
not
right right like like what happens like
besides
like they're very mad about like the
science continually
updating itself which is kind of exactly
what
science is supposed to do so i think
maybe a good understanding of the
integration with
um philosophy and you know the evolution
of science as
as a a thing a non-equilibrium steady
state that's continually being updated
like as a system i think that that might
um you know if it's pushed into
like the mainstream it might enable
people to be more accepting of
science when it changes
thomas thanks a lot blue

yeah um i think
it's it's very true that at least the
way we see it now if you
start talking about questions like is
gender real or not
that it has very um big impacts on
policy and how we're going to change
things
um and i don't know maybe this is a
typical instrumentalist thing to say but
i wonder if it should
matter as much if it's not just
we're seeing certain patterns of say
injustice and
we need to address
these patterns of injustice and try to
create policies
that remove that injustice regardless
of whether the
categories that um
we use in describing the population that
is affected by the injustice
latches on to something real or not
i don't think that really matters that
much it's
it's sort of the moment when you get
into
the philosophical debate about gender
when you're trying to
um discuss policy i think you're doing
policy making wrong
i think that you should be looking at
what
is going on is there injustice
in this case we can say yes there is and
how can we tackle that when you get into
the philosophical debate you're doing
something else

that's also important but not for
political policy making um
and i think in some ways um what
um i think yeah i think stephen said
that
um about trying to legitimize certain
um parts of research
by relating it to brain processes
like oh the thing i'm doing is only
really legitimate
if i can point at the brain and say like
these things are doing the thing i'm
talking about
um i think you see a similar thing
with um the
neurodiversity
groups where people
latch onto this scientific framework
of oh well autism is all in the brain so
it's real so now you have to take me
seriously
and i think that's in some sense
it's nice to see that you know these
things are being legitimized and people
are taking it more seriously
in another sense it's not good because
we're restricting ourselves very much to
a very particular
idea of what counts as real only if we
find it in the brain
and as we've just been talking about
just looking at the brain if you're
trying to
understand a broader picture of what an
agent in an environment is doing
is um very narrow um
yeah that that was it
great thanks so much for sharing that i

just wanted to
kind of build a little bit on that um
you said like whether we think
x is real there's so many issues
yesterday today tomorrow is x real
it becomes real when it has impact on
policy and what's so cool about active
inference is
we even use that term policy to talk
about how
things matter and we use the instruments
of these different models the specific
ones that we're talking about to talk
about how things bear upon
policy and so the two things or two of
the things that active inference can
help us do is first off it helps us
separate our preference
from our affordances from observations
from our policy
because it's really easy when talking
about things that really matter climate
change
all these other things it gets quite
blurred with what people's preferences
as far as their short-term affordances
long-term teleos um and then also it
gives us a forum
and a shared understanding and
vocabulary for actually having that
realism instrumentalism debate blue to
what you said about this evolving state
of science developing
state of science um you can veer off one
side
and say well it's real today and what
they say is real tomorrow
so maybe a realist stance lends itself

towards kind of in-group thinking
because if you're being told if the
party tells you it's real today
and the party tells you something new
next year you better be a realist about
that
at least in terms of policy versus if we
take a purely deconstructionist it's all
instrumentalist
then some people are going to say wait
but if it's not real like
i mean our virus is real we hope so so
what is real
and so that's the kind of questions that
we can have a discussion
that values people's perspective
modes of communication modes of thinking
being different roles
in complex adaptive systems together so
really awesome discussion we're gonna
land it on time but again
thanks everyone for participating if
you're listening you're also
welcome to participate in a future
discussion or in
any other activity of actin flab and
just
find us at activeinference.org and go
from there ns and thomas
we look forward to your future research
and having you back for any
conversations you'd ever like